

PC Score — Eigen values $\lambda_1, \lambda_2, \lambda_3$

* features ^{must} include

	<u>F_1</u>	<u>F_2</u>	<u>F_3</u>	<u>value</u>
$H \neq 1$	.	.	.	=
$H \neq 2$	.	.	.	

$$N = \frac{\vec{v}}{|\vec{v}|}$$

Eigenvektor:

Normal

$$\begin{pmatrix} V_1 \\ V_2 \\ V_3 \end{pmatrix}$$

$$P = (v_1 \ v_2 \ v_3) \begin{pmatrix} x - \bar{x} \\ y - \bar{y} \end{pmatrix} \quad \text{diagonal}$$

x	y	z	$x - \bar{x}$	$y - \bar{y}$	$z - \bar{z}$	Deviation		
						$(x - \bar{x})^2$	$(y - \bar{y})^2$	$(z - \bar{z})^2$
$\bar{x}$	$\bar{y}$	$\bar{z}$				Σ	Σ	Σ

$$\sigma_x = \frac{\sum (x - \bar{x})^2}{N-1}$$

$$s_y = \frac{\sum (y - \bar{y})^2}{N-1}$$

$$S_z = \frac{\sum (z - \bar{z})^2}{N-1}$$

$$s_{xy} = \frac{\sum xy}{n-1}$$

$$\frac{\sum xz}{n-1}$$

$$\sigma_{yz} = \frac{\sum yz}{n-1}$$

Co-variance matrix C

$$\begin{pmatrix} \sigma_x & \sigma_{xy} & \sigma_{xz} \\ \sigma_{xy} & \sigma_y & \sigma_{yz} \\ \sigma_{xz} & \sigma_{yz} & \sigma_z \end{pmatrix}$$

① Eigen values $|C - \lambda I| = 0$

Select greater 1-

$\lambda_1, \lambda_2, \lambda_3$

$$I = \frac{h_1}{h_1 + h_2 + h_3} =$$

$\lambda_1 \Rightarrow$ Eigenvektoren $\begin{pmatrix} a \\ b \\ c \end{pmatrix}$

Normal $|M| = \sqrt{a^2 - b^2 + c^2}$ $\begin{pmatrix} c \\ \vdots \end{pmatrix}$

$$v_1 = \frac{a}{m_1}$$
$$v_2 = \frac{b}{m_1}$$
$$v_3 = \frac{c}{m_1}$$

$$\rightarrow P(\underline{v_1} + \underline{v_2} + \underline{v_3}) \rightarrow$$

x, y, z, T

$$\begin{vmatrix} \sigma_x & \sigma_{xy} & \sigma_{xz} & \sigma_{xt} \\ \tau & \tau & \tau & \tau \end{vmatrix}$$

$$\sigma_{xy} = \frac{\sum (x - \bar{x})(y - \bar{y})}{N - 1}$$

$$\begin{pmatrix} \bar{\sigma}_x & \sigma_{xy} & \sigma_{xz} & \sigma_{xt} \\ \sigma_{xy} & \sigma_y & \sigma_{yz} & \sigma_{yt} \\ \sigma_{xz} & \sigma_{yz} & \sigma_z & \sigma_{zt} \\ \sigma_{xt} & \sigma_{yt} & \sigma_{zt} & \sigma_t \end{pmatrix}$$

$$\sigma_{xy} = \frac{\sum (x - \bar{x})(y - \bar{y})}{N-1}$$

$$\sigma_{xx} = \frac{\sum (x - \bar{x})^2}{N-1}$$

5. Step 4: Finding the Principal Component (Eigenvector)

For the largest eigenvalue $\lambda_1 = 6.53$, solve $(C - 6.53I)v = 0$:

$$\begin{bmatrix} 2.5 - 6.53 & 3.0 \\ 3.0 & 4.3 - 6.53 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} -4.03 & 3.0 \\ 3.0 & -2.23 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

From the first row: $-4.03v_1 + 3v_2 = 0 \Rightarrow v_2 \approx 1.34v_1$. Normalizing the vector $\begin{bmatrix} 1 \\ 1.34 \end{bmatrix}$ gives our First Principal Component:

$$v_1 \approx \begin{bmatrix} 0.6 \\ 0.8 \end{bmatrix}$$

$$EV = \begin{pmatrix} 1 \\ 1.34 \end{pmatrix}$$

$$|EV| = \sqrt{1^2 + 1.34^2} = 1.67$$

$$\begin{pmatrix} 0.6 \\ 0.8 \end{pmatrix} \frac{1}{1.67}$$

$$\begin{pmatrix} -0.8 \\ 0.6 \end{pmatrix}$$

$$\begin{pmatrix} -4.03 & 3 \\ 3 & -2.23 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$-4.03v_1 + 3v_2 = 0 \quad 3v_2 = 4.03v_1$$

$$3v_1 - 2.23v_2 = 0$$

$$2.23v_2 = 3v_1$$

$$v_2 = \frac{3}{2.23}v_1$$

$$v_2 = 1.34v_1 \quad v_1 = 1 \quad v_2 = 1.34$$

$$V = \begin{pmatrix} 0.6 \\ 0.8 \end{pmatrix}$$

Central diff

	X	Y	$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x})^2$	$(y - \bar{y})^2$	$(x - \bar{x})(y - \bar{y})$
	-1	-2	-1	-2	1	4	2
	2	3	2	3	4	9	6
	3	1	3	1	9	1	3
	0.7	+2	0.7	+2	0.49	4	1.4
			$\sum (x - \bar{x})^2$	$\sum (y - \bar{y})^2$	$\sum (x - \bar{x})(y - \bar{y})$		

① $\bar{x}, \bar{y}$

$V(x) = \frac{\sum (x - \bar{x})^2}{N-1}$, $Vary(\frac{\sum (y - \bar{y})^2}{N-1})$; Co-var: $\frac{\sum (x - \bar{x})(y - \bar{y})}{N-1}$

variance (x,y)

$$v^{(n)} = \frac{1}{\sqrt{N-1}}, \quad v^{(1)} = \frac{1}{\sqrt{N-1}}$$

$$v^{(1)}$$

Co-variance (

$$\begin{pmatrix} \text{Var}(x) & \text{Co variance}(x,y) \\ \text{Co variance}(x,y) & \text{Var}(y) \end{pmatrix} \Rightarrow C$$

Eigen value

Suppose

$$(C - \lambda I) = 0$$

$$\begin{vmatrix} 2-\lambda & 3 \\ 3 & 2-\lambda \end{vmatrix} = 0 \quad |C - \lambda I| = 0$$

$$(2-\lambda)^2 - 9 = 0$$

$$(2-\lambda+3)(2-\lambda-3) = 0$$

$$\lambda_1 = 5$$

$$-\lambda - 1 = 0$$

$$\lambda = 1$$

$$\lambda_2 = 1$$

$$\lambda_1 \quad \frac{5}{6} = 0.82\%$$

$$\lambda_2 \quad \frac{1}{6} = 0.18\%$$

$$\begin{pmatrix} v_1 & v_2 \\ -3 & 3 & 0 \\ 3 & -3 & 0 \end{pmatrix}$$

Eigen vector

$$3v_1 - 3v_2 = 0$$

$$v_1 = v_2$$

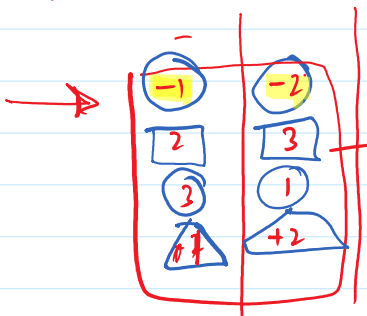
Normal

$$\sqrt{1^2 + 1^2}$$

$$= \sqrt{2}$$

$$PC_1 = \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix}$$

PC1



$$\left. \begin{aligned} -1 \times \frac{1}{\sqrt{2}} + -2 \times \frac{1}{\sqrt{2}} &= -\frac{3}{\sqrt{2}} \\ 2 \times \frac{1}{\sqrt{2}} + 3 \times \frac{1}{\sqrt{2}} &= \frac{5}{\sqrt{2}} \\ 3 \times \frac{1}{\sqrt{2}} + 1 \times \frac{1}{\sqrt{2}} &= \frac{4}{\sqrt{2}} \\ 1 \times \frac{1}{\sqrt{2}} + 2 \times \frac{1}{\sqrt{2}} &= \frac{3}{\sqrt{2}} \end{aligned} \right\}$$

λ_1 — PC₁

λ_2 — PC₂

Highest

λ_1

PC₁

PC₁

low.

λ_2

PC₂

① Affine Transformations

Matrix

① Scaling $\begin{pmatrix} K & 0 \\ 0 & K \end{pmatrix}$ $(x, y) \rightarrow (Kx, Ky)$

② Stretching $\begin{pmatrix} K_1 & 0 \\ 0 & 1 \end{pmatrix}$ $(x, y) \xrightarrow{K_1 \text{ I.L. } x\text{-axis}} (K_1x, y) \rightarrow \text{Along } x\text{-axis}$

③ SHEARIN $\begin{pmatrix} 1 & 0 \\ 0 & K_2 \end{pmatrix}$ $(x, y) \xrightarrow{K_2 \text{ I.L. } y\text{-axis}} (x, K_2y) \Rightarrow \text{Along } y\text{-axis}$

Two way stretch $\begin{pmatrix} K_1 & 0 \\ 0 & K_2 \end{pmatrix}$ $(x, y) \Rightarrow (K_1x, K_2y)$ scaling $K_1 = K_2 \Rightarrow K$



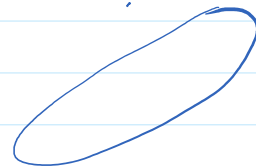
$$\begin{pmatrix} 1 & Hx \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = (x + Hx \cdot y, y)$$

$$\begin{pmatrix} 1 & 0 \\ Hy & 1 \end{pmatrix} \Rightarrow \text{Along or Invariant in } y\text{-axis}$$

$$(4, 6), (3, 8), (-1, 2) \quad \left| \begin{pmatrix} K & 0 \\ 0 & K \end{pmatrix} \begin{pmatrix} 4 & 3 & -1 \\ 6 & 8 & 2 \end{pmatrix} \right|$$

Function

$$\frac{(x-2)^2}{4} + \frac{(y-3)^2}{4} = 1$$



Shear $x\text{-axis I.L. } 2$

$$\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x' \\ y' \end{pmatrix}$$

$$x'^2 + 4y'^2 - 4x'y' + y'^2 + 6y' + 9 - 4 = 0$$

$$\boxed{x'^2 - 4x'y' + 5y'^2} + 0x' + 6y' + 5 = 0$$

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$$

$$A=1, B=-4, C=5$$

$$x + 2y = x' \Rightarrow x + 2y' = x'$$

$$y = y' \Rightarrow \begin{cases} x = x' - 2y' \\ y = y' \end{cases}$$

$$B^2 - 4AC = 16 - 4(1)(5) = -4$$

$$B^2 - 4AC < 0 \quad \text{Ellips}$$

$$> 0 \quad \text{Hyperbola}$$

= 0 parabola

Vector spaces

Example 1: Let $V = R^2 = \{(x, y); x, y \in R\}$, prove that V is a vector space under the usual operations of addition and scalar multiplication defined by:

$$\vec{u} + \vec{v} = (u_1, u_2) + (v_1, v_2) = (u_1 + v_1, u_2 + v_2)$$

$$k\vec{u} = k(\vec{u}_1, \vec{u}_2) = (k\vec{u}_1, k\vec{u}_2)$$

$$k\vec{u} = k(\vec{u}_1, \vec{u}_2, \vec{u}_3) = (k\vec{u}_1, k\vec{u}_2, k\vec{u}_3)$$

Example 3: Let $V = R^2$, under the usual operations of addition defined by:

$$\vec{u} + \vec{v} = (\vec{u}_1, \vec{u}_2) + (\vec{v}_1, \vec{v}_2) = (\vec{u}_1 + \vec{v}_1, \vec{u}_2 + \vec{v}_2)$$

And if k is any scalar number, then define

$$k\vec{u} = k(\vec{u}_1, \vec{u}_2) = (k\vec{u}_1, 0)$$

The addition operation is standard one on R^2 , but the scalar multiplication is not.

Check whether V is vector space or not?

Question 3: [10 marks]

Find all the eigenvalues of the given matrix. Also, find eigenvector for **negative** eigenvalue of the matrix

$$A = \begin{bmatrix} 2 & -3 & 0 \\ 2 & -5 & 0 \\ 1 & 0 & 3 \end{bmatrix}$$

Question 4: [6+4 marks]

i) Let $V = R^3 = \{(x, y, z); x, y, z \in R\}$ be a set of ordered triplex of real number with the following operations of addition and scalar multiplication:

$$\vec{u}_1 + \vec{u}_2 = (x_1, y_1, z_1) + (x_2, y_2, z_2) = (x_1 + x_2 - 2, y_1 + y_2, z_1 + z_2 + 3)$$

$$k(x_1, y_1, z_1) = (1, ky_1, kz_1)$$

a. Compute $\vec{u}_1 + \vec{u}_2$ if $\vec{u}_1 = (1, -1, 3)$, $\vec{u}_2 = (-4, 2, -7)$.

b. Compute $k\vec{u}_1$ if $\vec{u}_1 = (2, 3, -5)$ and $k = -2$.

c. Find the zero vector of the set.

d. Find the additive inverse of the vector $(3, 2, 5)$.

e. Prove or disprove the property: $k(\vec{u}_1 + \vec{u}_2) = k\vec{u}_1 + k\vec{u}_2$.

f. Prove or disprove the property: $1\vec{u} = \vec{u}$.

ii) Let $V = R^2 = \{(x, y); x \in R, y \in R\}$ be a vector space under standard addition and scalar multiplication. Prove or disprove the following subsets are **subspaces**.

a. $W_1 = \{(x, y); 0 \leq x \leq 1, 0 \leq y \leq 1\}$

b. $W_2 = \{(x, 0); x \in R\}$

$$(x_1, y_1, z_1) + (x_2, y_2, z_2) = (\underline{x_1 + x_2 - 2}, \underline{y_1 + y_2}, \underline{z_1 + z_2 + 3})$$

$$12(x, y, z) = (1, 12y, 12z)$$

$$\left\{ \begin{array}{l} u + v = (0, 0, 0) \quad \checkmark \\ (x, y, z) + (a, b, c) = (0, 0, 0) \\ \underline{x + a - 2 = 0} \quad a = 2 - x \\ y + b = 0 \quad b = -y \\ z + c + 3 = 0 \quad c = -3 - z \end{array} \right. \quad u + 0 = u$$

$$u(x, y, z), \quad \boxed{0(a, b, c)} \Rightarrow \quad (u + \vec{0} = \vec{u})$$

$$u + 0 = u$$

$$(x, y, z) + (a, b, c) = (x, y, z) \quad (0(2, 0, -3))$$

$$\begin{array}{l} x + a - 2 = x \Rightarrow a = 2 \\ y + b = y \Rightarrow b = 0 \\ z + c + 3 = z \Rightarrow c = -3 \end{array}$$

$$(x, y, z) + (2, 0, -3) = (x + 2 - 2, y + 0, z - 3 + 3) = (x, y, z)$$

$$(x_1 + x_2 - 2, y_1 + y_2, z_1 + z_2 + 3)$$

$$0(0, 0, 0)$$

$$\vec{u} + \vec{0} = \vec{u}$$

Zero vector

$$u(x, y, z), \quad 0(a, b, c)$$

$$(x, y, z) + (a, b, c) = (x, y, z)$$

$$0(2, 0, -3) \quad (x + a - 2, y + b, z + c + 3) = (x, y, z)$$

$$\begin{array}{l} x + a - 2 = x \quad a = 2 \\ y + b = y \quad b = 0 \\ z + c + 3 = z \quad c = -3 \end{array}$$

$$(-3, -2, -5)$$

$$(3, 2, 5)$$

$$(x, y, z)$$

$$\underline{v} + \underline{w} = \underline{\vec{0}}$$

$$(3, 2, 5) + (-3, -2, -5)$$

$$(0, 0, 0)$$

$$u + v = 0$$

$$(2, 2, 5) + (x, y, z) - (2, 0, -3)$$

$$(3, 2, 5) + (x, y, z) = (2, 0, -3)$$

$$(3+x-2, 2+y, 5+z+3) = (2, 0, -3)$$

$$(1+x, 2+y, 8+z) = (2, 0, -3)$$

$$1+x = 2$$

$$2+y = 0$$

$$8+z = -3$$

$$x = 1$$

$$y = -2$$

$$z = -11$$

$$\begin{pmatrix} 3 & 2 & 5 \\ 1 & -2 & -11 \end{pmatrix}$$

$$\begin{pmatrix} 4-2 & 0 & 5-11+3 \\ 2 & 0 & -3 \end{pmatrix}$$

$$O = (2, 0, -3)$$



$$(3, 2, 5) + (x, y, z) = (2, 0, -3)$$

$$(3+x-2, 2+y, 5+z+3) = (2, 0, -3)$$

$$(x+1, y+2, z+8) = (2, 0, -3)$$

$$x+1 = 2$$

$$y+2 = 0$$

$$z+8 = -3$$

$$x = 1$$

$$y = -2$$

$$z = -11$$

$$\begin{pmatrix} 1 & -2 & -11 \\ 3 & 2 & 5 \end{pmatrix}$$

$$(2, 0, -3)$$

Question 3: [10 marks]

Find all the eigenvalues of the given matrix. Also, find eigenvector for *negative* eigenvalue of the matrix

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- i) Let $V = R^3 = \{(x, y, z), x, y, z \in R\}$ be a set of ordered triples of real number with the following operations of addition and scalar multiplication:

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$$k(x_1, y_1, z_1) = (1, ky_1, kz_1)$$

- a. Compute $\vec{u}_1 + \vec{u}_2$ if $\vec{u}_1 = (1, -1, 3)$, $\vec{u}_2 = (-4, 2, -7)$.
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 e. Prove or disprove the property: $k(\vec{u}_1 + \vec{u}_2) = k\vec{u}_1 + k\vec{u}_2$.
 f. Prove or disprove the property: $1\vec{u} = \vec{u}$
- ii) Let $V = R^2 = \{(x, y), x \in R, y \in R\}$ be a vector space under standard addition and scalar multiplication. Prove or disprove the following subsets are *subspaces*:

a. $W_1 = \{(x, y) : 0 \leq x \leq 1, 0 \leq y \leq 1\}$

b. $W_2 = \{(x, 0) : x \in R\}$

$(x, y) + (x_2, y_2)$
 $(x_1 + x_2, y_1 + y_2)$
 $(x_1 + x_2 - 2, y_1 + y_2 - 1)$ *unusual*

$K(x_1, y_1)$
 (Kx, Ky)
 (K^2x, Ky)

$K(\vec{u}_1 + \vec{u}_2)$

$K\vec{u}_1 = K(x_1, y_1, z_1)$
 $= (1, Ky_1, Kz_1)$

$K\vec{u}_2 = K(x_2, y_2, z_2)$
 $= (1, Ky_2, Kz_2)$

$u_1 = (x_1, y_1, z_1)$, $u_2 = (x_2, y_2, z_2)$

$u_1 + u_2 = (x_1 + x_2 - 2, y_1 + y_2, z_1 + z_2 + 3)$

$K(u_1 + u_2) = (1, K(y_1 + y_2), K(z_1 + z_2 + 3))$

$= (1, Ky_1 + Ky_2, Kz_1 + Kz_2 + 3)$

$Ku_1 + Ku_2 = (1, Ky_1, Kz_1) + (1, Ky_2, Kz_2)$

$= (1+1-2, Ky_1 + Ky_2, Kz_1 + Kz_2 + 3)$

$Ku_1 + Ku_2 = (0, K(y_1 + y_2), K(z_1 + z_2 + 3))$

$1\vec{u} = \vec{u}$

$1(x, y, z) = (1, y, z)$ ✓

disprove

① $u = (1, x)$

$u + \dots = \dots$

$0 = (1, b)$

$(1, x) + (1, b) = (1, x)$

$(1, x+b) = (1, x)$

$x+b=x$ $b=0$

Example 8: Determine whether the set of all pairs of real numbers of the form

$(1, x)$ with the operations

$(1, y) + (1, y') = (1, y + y')$

$k(1, y) = (1, ky)$

is a vector space or

$0(1, 0)$

$$(1, 0) + (0, 1) = (1, 1)$$

$$(1, 0) + (0, 1) = (1, 1)$$

$$x+b=x \quad b=0$$

$$(1, x) + (1, y) = (1, 0)$$

$$(1, -x)$$

$$1, \quad x+y=0 \\ y=-x$$

$$(1, 5) + (1, 0) = (1, 5)$$

$$(1, -5)$$

$$(1, 5) + (1, b) = (1, 0)$$

$$(1, 5+b) = (1, 0) \\ 5+b=0 \\ b=-5$$

① $P(x)$: degree is greater than 3 usual

$$2t^3 + 4t^2 + 3t + 5,$$

$$2t^4 - 3t^2 + 4t + 5$$

$$⊕ \quad P(x) \geq 3$$

$$\text{Scalar } k \quad P(x) = \geq 3$$

$$P+Q = Q+P$$

$$(P+Q) + (R) = P + (Q+R)$$

○ Polynomial

degree Less Than

$$P(x) \quad 2t^3 + 4t^2 + 3t + 5$$

$$Q(x) \quad -2t^3 + 5t^2 + 6t - 7$$

* All Polynomials have degree greater than or equal to 'n' will not make a vector space

for degree $\leq n$, all of them will make a vector space.

less 5

$$2t^4 + 3t^2 + t,$$

$$2t^5 - 3t^3 + 4t + 5$$

$$-2t^4 - 3t^2 - t$$

$$0 \quad 0 \quad 0$$

$P(x)$ degree — highest power
 $\geq n$ let $n=5$

$$P_1(x) = 2x^5 - 4x^4 + 7x + 8$$

$$P_2(x) = -2x^5 - 3x^4 + 8x^2 + 9$$

$$P_1 + P_2 = \boxed{-7x^4 + 8x^2 + 7x + 17}$$

→ is not a vector space

✓

$$\textcircled{\leq 5}$$

$\leq n$ (always a vector space)

$V =$ greater than $n \neq 5$

$$+ \left\{ \begin{array}{l} 2t^5 + 3t^4 - 3t + 4 \\ -2t^5 + 4t^4 - t^2 + 5 \end{array} \right\}$$

$$\boxed{7t^4 - t^2 - 3t + 9}$$

Cubic + quadratic

ord = 5

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$$k(x_1, y_1, z_1) = (1, ky_1, kz_1)$$

- Compute $\vec{u}_1 + \vec{u}_2$ if $\vec{u}_1 = (1, -1, 3)$, $\vec{u}_2 = (-4, 2, -7)$.
- Compute $k\vec{u}_1$ if $\vec{u}_1 = (2, 3, -5)$ and $k = -2$.
- Find the zero vector of the set.
- Find the additive inverse of the vector $(3, 2, 5)$.
- Prove or disprove the property: $k(\vec{u}_1 + \vec{u}_2) = k\vec{u}_1 + k\vec{u}_2$.
- Prove or disprove the property: $1\vec{u} = \vec{u}$.

- ii) Let $V = R^2 = \{(x, y), x \in R, y \in R\}$ be a vector space under standard addition and scalar multiplication. Prove or disprove the following subsets are **subspaces**.

a. $W_1 = \{(x, y) : 0 \leq x \leq 1, 0 \leq y \leq 1\}$

b. $W_2 = \{(x, 0) : x \in R\}$ $\rightarrow$ Subspace

$u+v \in R$
 $kv \in R$

$x \ W_1$

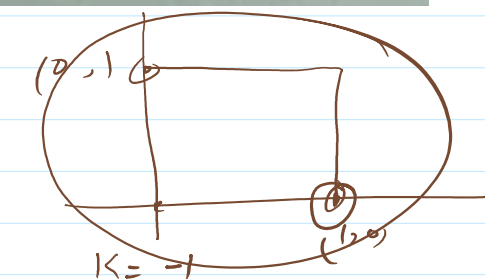
~~1, 1, 1~~

$(1, 0) + (0, 1) = (1, 1)$ ✓

$K(1, 1) = (-1, -1)$

$K=2, \quad 2(1, 1) = (2, 2) \notin W_1$

$(x, 0) ; \underline{x \in R}$



Closure of multipli.

$u_1 = (x_1, 0) \quad u_1 + u_2 = (\underline{x_1 + u_2}, 0) \in R$
 $u_2 = (x_2, 0)$

$K u_1 = K(x_1, 0)$
 $= (\underline{Kx_1}, 0)$

$\forall x_1 \in R, \underline{Kx_1 \in R}$

2. Use Theorem 4.2.1 to determine which of the following are subspaces of M_n .

- (a) The set of all diagonal $n \times n$ matrices.
- (b) The set of all $n \times n$ matrices A such that $\det(A) = 0$.
- (c) The set of all $n \times n$ matrices A such that $\text{tr}(A) = 0$.
- (d) The set of all symmetric $n \times n$ matrices.
- (e) The set of all $n \times n$ matrices A such that $A^T = -A$.
- (f) The set of all $n \times n$ matrices A for which $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.
- (g) The set of all $n \times n$ matrices A such that $AB = BA$ for some fixed $n \times n$ matrix B .

$$A = \begin{pmatrix} 1 & -1 \\ 2 & 3 \end{pmatrix}$$

$$B = \begin{pmatrix} 2 & 3 \\ -1 & 1 \end{pmatrix}$$

$$\Rightarrow A+B = \begin{pmatrix} 1 & -1 \\ 2 & 3 \end{pmatrix} + \begin{pmatrix} 2 & 3 \\ -1 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 3 & 2 \\ 1 & 4 \end{pmatrix}$$

$$\#2 \ B = \begin{pmatrix} 4 & 6 \\ -2 & 2 \end{pmatrix}$$

$$= 8 - (-12)$$

$$= 8 + 12 = 20$$

$$\begin{pmatrix} 1 & -1 & 0 \\ 2 & 3 & 0 \\ 0 & 5 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 & -1 & 0 \\ 0 & 5 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

$k_2 =$

$$12 - 2 = 10$$

$$\begin{pmatrix} 1 & 4 & 0 \\ 3 & 2 & 0 \\ 0 & -10 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 4 & 0 \\ 0 & -10 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

$$A\mathbf{x} = \mathbf{0}$$

$$\begin{pmatrix} -2 & 1 \\ 3 & 5 \end{pmatrix} + \begin{pmatrix} -2 & 3 \\ 1 & 5 \end{pmatrix} = \begin{pmatrix} -4 & 4 \\ 4 & 10 \end{pmatrix}$$

$$\begin{pmatrix} -4 & 4 \\ 4 & 10 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$x_2 = 0$

$$\begin{pmatrix} -2 & 1 & 0 \\ 3 & 5 & 0 \\ 1 & 6 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 6 & 0 \\ 3 & 5 & 0 \end{pmatrix}$$

$$\#2 \ B = \begin{pmatrix} 1 & 6 & 0 \\ 1 & 1 & 0 \end{pmatrix}$$

nr

$$R_2 - 3R_1 \begin{pmatrix} 1 & 6 & 0 \\ 0 & -13 & 0 \end{pmatrix}$$

$$\begin{pmatrix} x_1 & x_2 \\ 1 & 6 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

$x_2 = 0, x_1 = 0$

Question 3: [10 marks]
Find all the eigenvalues of the given matrix. Also, find eigenvector for *negative* eigenvalue of the matrix

$$A = \begin{bmatrix} 2 & -3 & 0 \\ 2 & -5 & 0 \\ 1 & 0 & 3 \end{bmatrix}$$

Question 4: [6+4 marks]

i) Let $V = R^3 = \{(x, y, z), x, y, z \in R\}$ be a set of ordered triples of real number with the following operations of addition and scalar multiplication:

$$\vec{u}_1 + \vec{u}_2 = (x_1, y_1, z_1) + (x_2, y_2, z_2) = (x_1 + x_2 - 2, y_1 + y_2, z_1 + z_2 + 3)$$

$$k(x_1, y_1, z_1) = (1, ky_1, kz_1)$$

- Compute $\vec{u}_1 + \vec{u}_2$ if $\vec{u}_1 = (1, -1, 3)$, $\vec{u}_2 = (-4, 2, -7)$.
- Compute $k\vec{u}_1$ if $\vec{u}_1 = (2, 3, -5)$ and $k = -2$.
- Find the zero vector of the set.
- Find the additive inverse of the vector $(3, 2, 5)$.
- Prove or disprove the property: $k(\vec{u}_1 + \vec{u}_2) = k\vec{u}_1 + k\vec{u}_2$.
- Prove or disprove the property: $1\vec{u} = \vec{u}$.

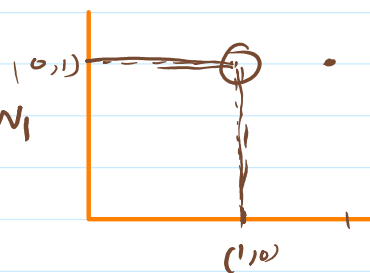
ii) Let $V = R^2 = \{(x, y), x \in R, y \in R\}$ be a vector space under standard addition and scalar multiplication. Prove or disprove the following subsets are *subspaces*.

- $W_1 = \{(x, y) : 0 \leq x \leq 1, 0 \leq y \leq 1\}$
- $W_2 = \{(x, 0) : x \in R\}$ is a subspace

a)

$$(1, 0) + (1, 1) = (2, 1) \notin W_1$$

Not a subspace



$$W_2 = \{(x, 0), x \in R\}$$

$$(-3, 0), (5, 0)$$

$$(x_1, 0) + (x_2, 0) = (x_1 + x_2, 0 + 0) = (x_1 + x_2, 0)$$

$$-3 + 5, 0 + 0 = (2, 0) \in W_2$$

$$x_1 \in R, x_2 \in R, x_1 + x_2 \in R$$

$$K(x, 0) = (Kx, 0)$$

Closure property holds

$$K(x, 0) = (Kx, 0)$$

$\downarrow \in \mathbb{R}$

... property holds

$$-8(4, 0) = (-32, 0)$$

$$(32, 0) \in \mathbb{R}$$

$$+8(32, 0) = (32, 0)$$

$\downarrow \in \mathbb{R}$

Q4. [Marks: 3 + 5 + 2 = 10]

Consider the following equation of circle

$$(x+2)^2 + (y-1)^2 = 4$$

a) Translate this circle by vector (2, 3).

b) Find equation of translated circle under the effect of stretching parallel to y-axis by factor 3.

c) Sketch the original and the images of circle obtained in part 'a' and 'b'.

$$A = \begin{bmatrix} 1 & 0 \\ 0 & k \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$$

Q#4 Given eq. of circle is

$$(x+2)^2 + (y-1)^2 = 4 \quad \text{--- (1)}$$

(a) Translation:

By definition; we have

$$T(\vec{x}) = \vec{x} + \vec{a}$$

Then $\vec{x} = \begin{bmatrix} x \\ y \end{bmatrix}$, and $\vec{a} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$ (given)

So

$$T \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x+2 \\ y+3 \end{bmatrix}$$

$$\Rightarrow x' = x + 2 \quad ; \quad y' = y + 3$$

$$\Rightarrow x' - 2 = x \quad ; \quad y' - 3 = y$$

$$\Rightarrow \boxed{x = x' - 2} \quad ; \quad \boxed{y = y' - 3}$$

putting values in eq (1), we have

$$(x' - 2 + 2)^2 + (y' - 3 - 1)^2 = 4$$

$$\boxed{(x')^2 + (y' - 4)^2 = 4} \quad \text{--- (2)}$$

$$x''^2 + \left(\frac{1}{3}y'' - 4\right)^2 = 4$$

$$x''^2 + \left(\frac{y'' - 12}{3}\right)^2 = 4$$

$$9(x'')^2 + \frac{(y'' - 12)^2}{3} = 36$$

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} 2 \\ 3 \end{pmatrix}$$

$$x' = x + 2$$

$$x' - 2 = x$$

$$y' = y + 3$$

$$y' - 3 = y$$

$$\begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} x'' \\ y'' \end{pmatrix}$$

$$x' = x''$$

$$3y' = y''$$

$$y' = \frac{1}{3}y''$$

$$\begin{vmatrix} T = x + a \\ 0 & 1 \end{vmatrix}$$

$$9(x'')^2 + (y'' - 12)^2 = 36$$

$$\frac{(x'' - 0)^2}{4} + \frac{(y'' - 12)^2}{36} = 1$$

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$$

$$k_1 \begin{pmatrix} a \\ b \\ c \end{pmatrix} + k_2 \begin{pmatrix} d \\ e \\ f \end{pmatrix} + k_3 \begin{pmatrix} g \\ h \\ i \end{pmatrix} = \begin{pmatrix} A \\ B \\ C \end{pmatrix}$$

$$\begin{pmatrix} 2 & 1 & -1 & 0 \\ 3 & 2 & 2 & 0 \\ 4 & 2 & 3 & 0 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix}$$

$-R_1 + R_2$

$$\begin{pmatrix} 1 & 1 & 3 & 0 \\ 3 & 2 & 2 & 0 \\ 4 & 2 & 3 & 0 \end{pmatrix}$$

① Linearly Independent

↓
② Span Also

$$\begin{matrix} R_2 - 3R_1 \\ R_3 - 4R_1 \end{matrix} \begin{pmatrix} 1 & 1 & 3 & 0 \\ 0 & -1 & -7 & 0 \\ 0 & -2 & -9 & 0 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix}$$



WhatsApp
Image 20...

Let $V = \mathbb{R}^3 = \{(x, y, z), x, y, z \in \mathbb{R}\}$ be a set of triples of real number with the following addition and scalar multiplication operations:

$$\vec{u}_1 + \vec{u}_2 = (x_1, y_1, z_1) + (x_2, y_2, z_2) = (x_1 + x_2 + 1, y_1 + y_2 - 1, z_1 + z_2)$$

$$k(x_1, y_1, z_1) = (0, ky_1, kz_1).$$

- Compute $\vec{u}_1 + \vec{u}_2$ if $\vec{u}_1 = (4, -1, 8)$, $\vec{u}_2 = (3, 2, -5)$.
- Compute $k\vec{u}_1$ if $\vec{u}_1 = (4, -1, 8)$ and $k = 2$.
- Find the zero vector of the set.
- Find the additive inverse of the vector $(2, 3, 4)$.
- Prove or disprove the property: $k(\vec{u}_1 + \vec{u}_2) = k\vec{u}_1 + k\vec{u}_2$.
- Prove or disprove the property: $1\vec{u} = \vec{u}$.

ii) [4 pts] Let $V = \mathbb{R}^2 = \{(x, y), x \in \mathbb{R}, y \in \mathbb{R}\}$ be a vector space under standard addition and scalar multiplication.

Prove/Disprove the following subsets are *subspaces*.

- $W_1 = \{(x, y): x \in \mathbb{R}, y = -2x\}$
- $W_2 = \{(x, y): x = y^2, y \in \mathbb{R}\}$

Question 4: [4+6pts]

- Let $P_2(x)$ be a vector space of all 2nd degree polynomials.
Show that $p = x^2 - 1$, $q = x^2 + x + 2$ and $r = -4x^2 - 2x - 2$ are linearly dependent vectors of $P_2(x)$.

- Show that the set of vectors $S = \{(1, 2, 3), (1, 4, 6), (2, -3, -5)\}$ span $\mathbb{R}^3$.
Use your result to express $v = (9, 1, 0)$ as a linear combination of the given vectors

①

$$K_1(x^2-1) + K_2(x^2+x+2) + K_3(-4x^2-2x-2) = 0x^2 + 0x + 0$$

$$K_1x^2 - K_1 + K_2x^2 + K_2x + 2K_2 - 4K_3x^2 - 2K_3x - 2K_3$$

$$K_1 + K_2 - 4K_3 = 0$$

$$K_2 - 2K_3 = 0$$

$$-K_1 + 2K_2 - 2K_3 = 0$$

$$\begin{pmatrix} 1 & 1 & -4 & 0 \\ 0 & 1 & -2 & 0 \\ -1 & 2 & -2 & 0 \end{pmatrix}$$

$$K_1 \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + K_2 \begin{pmatrix} 1 \\ 4 \\ 6 \end{pmatrix} + K_3 \begin{pmatrix} 2 \\ -3 \\ -5 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 & -4 & 0 \\ 0 & 1 & -2 & 0 \\ 0 & 3 & -6 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 & -4 & 0 \\ 0 & 1 & -2 & 0 \\ 0 & 1 & -2 & 0 \end{pmatrix}$$

$$\det A = \begin{vmatrix} 1 & 1 & 2 \\ 2 & 4 & -3 \\ 3 & 6 & -5 \end{vmatrix} \begin{vmatrix} 9 \\ 1 \\ 0 \end{vmatrix}$$

$$\begin{pmatrix} 1 & 1 & -4 & 0 \\ 0 & 1 & -2 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$1(-20+18) - 1(-10+9) + 2(12-12) = 0$$

$$-2 + 1 = -1 \neq 0$$

$$K_2 - 2K_3 = 0$$

$$K_2 = 2K_3$$

$$K_1 + 2t - 4t = 0$$

$$K_1 = 2t$$

It spans

* Independent vectors $\leftrightarrow$ spans
 $\det[A] \neq 0$

Basis

Shows

* linear combinations